MAT 3103: Computational Statistics and Probability
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Chapter 7[Sampling Technique]

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Sampling Technique

Definition: It is a technique to select a representative part of population units, where units are investigated to study the characteristics of population units.

Different Methods of Sampling are:

i) Simple random sampling, **ii)** Systematic sampling, **iii)** Circular systematic sampling, **iv)** Cluster sampling, **v)** Multistage sampling,**)** Stratified sampling etc.

Simple Random Sampling: Let there be N units in a population. We need to select a random sample of size n ($n \leq N$). The possible number of samples, without replacement, are ${}^{N}C_{n}$. If any of these ${}^{N}C_{n}$ samples are selected with equal probability $1 / {}^{N}C_{n}$, then the sampling is known as simple random sampling. In other words, If every unit of N units is selected with equal probability 1/ N, then the sampling is called simple random sampling.

Example: Prove by example,

(i) Sample mean is an unbiased estimate of Population mean
(ii) Sample variance is an unbiased estimate of Population variance.

Solution: Let us consider that, in a population there are N = 3 units. The unit values are x: 2, 4, 6. We can select a sample of size n =2. The possible number of samples, without replacement, are ${}^{3}C_{2} = 3$.

The samples and the characteristics of the samples are shown below:

Samples	**Sample Means,** $\bar{x} = \frac{1}{n}\Sigma x$	**Sample Variances,** $s^2 = \frac{1}{n-1}\ [\Sigma x^2 - \frac{(\Sigma x)^2}{n}]$
2, 4	$\bar{x}_1 = \frac{1}{n}\Sigma x = \frac{2+4}{2} = 3$	$s_1^2 = \frac{1}{2-1}[(2^2 + 4^2) - \frac{(2+4)^2}{2}] = 2$
2, 6	$\bar{x}_2 = \frac{2+6}{2} = 4$	$s_2^2 = \frac{1}{2-1}[(2^2 + 6^2) - \frac{(2+6)^2}{2}] = 8$
4, 6	$\bar{x}_3 = \frac{4+6}{2} = 5$	$s_3^2 = \frac{1}{2-1}[(4^2 + 6^2) - \frac{(4+6)^2}{2}] = 2$

Among the above 3 samples, if any one of the samples is selected with probability 1 /3, then the selected sample is a simple random sample. The population mean and the population variance of the above given population are as follows:

$$Population\ Mean,\ \bar{X} = \frac{1}{N}\Sigma X = \frac{1}{3}(2+4+6) = 4,$$

$$Population\ Variance, S^2 = \frac{1}{N-1}[\Sigma x^2 - \frac{(\Sigma x)^2}{N}]$$

$$= \frac{1}{3-1}[(2^2 + 4^2 + 6^2) - \frac{(2+4+6)^2}{3}] = 4$$

The sample mean is an unbiased estimate of population mean as mean of sample means is equal to the population mean as shown below:

Mean of sample means, E(X) = $\frac{1}{3}(3+4+5) = 4 = \bar{X}$ [Proved]

The sample variance is an unbiased estimate of population variance as mean of sample variances is equal to the population variance as shown below:

Mean of sample variances, E(S^2) = $\frac{1}{3}(2+8+2) = 4 = S^2$ [Proved]

Formula according to variance:

The **estimate of the variance of sample means** is given by

$$v(\bar{x}) = \frac{N-n}{Nn} s^2 .$$

The **estimated standard error of sample means** is also needed and it is given by

$$\text{s.e.}(\bar{x}) = \sqrt{v(\bar{x})} .$$

The total values of the population units can also be estimated. The **estimate of population total** is given by

$$\hat{X} = N\bar{x}$$

The estimate of variance of the estimate of population total is given by

$$v(\hat{X}) = N^2 v(\bar{x}) \text{ and } s.e.(\hat{X}) = \sqrt{v(\hat{X})}$$

Estimated Proportion: Let

N = Number of population units,

n= Number of sample units,

A = Number of units in the population possessing a particular character,

a = Number of sample units possessing that particular character,

P = **A /N** =Proportion of **population** units possessing that particular character,

P = a /n = Proportion of **sample** units possessing that particular character.

This **p is an unbiased estimate of P**. The **estimated variance of Proportion** is given by

$$V(p) = \frac{N-n}{N(n-1)} pq \text{ , } q = 1 - p$$

Example: The numbers of pharmacies in different regions of a city are, X: 12, 20, 5, 25, 10, 35, 8, 15, 20, 13, 20, 18, 8, 9, 24, 25, 15, 30, 18, 22, 25, 17, 30, 25, 18, 20, 22, 20. Select a random sample of 6 regions by (i) simple random sampling method, (ii) systematic sampling method, and by (iii) circular systematic sampling method.

1. Estimate mean number of pharmacies per region.
2. Estimate total number of pharmacies in the city.
3. Estimate standard error of estimated mean number of pharmacies.
4. Estimate standard error of estimated total number of pharmacies.
5. Find 95% confidence interval for mean number of pharmacies.

6. Is the average number of pharmacies per region 18?
7. Suggest a sample of size n to estimate a population mean with margin of error 0.5 at 95% level of confidence, where variance of the population observations is 6.35.
8. Estimate the proportion of regions in which there are less than 18 pharmacies.
9. Estimate the variance of the estimated proportion.
10. Suggest a sample size n to estimate a proportion 0.75 with margin of error 0.2 at 95% confidence.
11. Do you think that the proportion of regions in which there are less than 18 pharmacies is 0.6?

The observations are shown by assigning a serial number to each observation as:

Serial Number	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Observations (x)	12	20	5	25	10	35	8	15	20	13	20	18	8	9
Serial Number	15	16	17	18	19	20	21	22	23	24	25	26	27	28
Observations (x)	24	25	15	30	18	22	25	17	30	25	18	20	22	20

i) We have N = 28. We need to select a sample of size n = 6. Six regions are selected using Random Number Table. We can use any row or any column of the table. Let us use Column 1 of the Random Number Table. Here N =28, the last serial number is of two digits. So, we need to select a random numbers of two digits. The selected random numbers and the selected number of pharmacies of different regions are shown below:

Random Numbers	13	25	4	1	11	7
Pharmacies of selected regions	8	18	25	12	20	8

ii) Let $N = nk$, $k = N/n$. In our case k = 28/6 = 4.7 ~ 5. We have to select first observation from first k= 5 observations using Random Number Table. After that every $k^{th} = 5^{th}$ observation is selected. Five is a number of one digit, so we need to select a random number of one digit first. The selected random numbers and the selected observations are shown below: [using column 1 of random number table]

Random Numbers	3	8	13	18	23	28
Pharmacies of selected regions	5	15	8	30	30	20

iii) Here also k = N/n =4.7 ~ 5. First observation is selected from all observations using random number table. After that every $k^{th} = 5^{th}$ observation is selected moving through a circle. We have n = 28 observations and 28 is a number of two digits. So we need to select a random number of two digits. The selected random number and the selected observations are shown below: [Using column 1 of random number table]

Random Numbers	13	18	23	28	5	10
Pharmacies of selected regions	8	30	30	20	10	13

Calculation is from Circular Systematic Sampling

1. Estimate of mean, $\bar{x} = \frac{1}{n}\Sigma x = \frac{111}{6} = 18.5$

2. Estimate of total, $\hat{X} = N\bar{x} = 28 \times 18.5 = 518$

3. The standard error of estimate of mean is, $\text{s.e.}(\bar{x}) = \sqrt{v(\bar{x})} = \sqrt{12.56} = 3.54$.

The variance of sample mean is $v(\bar{x}) = \frac{N-n}{Nn} s^2 = \frac{28-6}{28\times 6} \times 95.9 = 12.56$

$$s^2 = \frac{1}{n-1}\left[\Sigma x^2 - \frac{(\Sigma x)^2}{n}\right] = \frac{1}{6-1}\left[2533 - \frac{(111)^2}{6}\right] = 95.9$$

4. The estimate of standard error of estimate of population total is given as follows:

$v(\hat{X}) = N^2 v(\bar{x}) = 28^2 \times 12.56 = 9847.04$ *and* $\text{s.e.}(\hat{X}) = \sqrt{v(\hat{X})} = \sqrt{9847.04} = 99.23$

5. 95% confidence limits are given as follows:

$$\hat{\bar{X}}_L = \bar{x} - t_{0.05,n-1}\ \text{s.e.}(\bar{x}) = 18.5 - 2.571 \times 3.54 = 9.4 \quad \textit{and}$$

$$\hat{\bar{X}}_U = \bar{x} + t_{0.05,n-1}\ \text{s.e.}(\bar{x}) = 18.5 + 2.571 \times 3.54 = 27.6$$

Here $t_{0.05,n-1}$ is the tabulated value of t at 5% level of significance with (n – 1) d.f.= 2.571

6. We need to test the null hypothesis , $H_0 : \mu = \mu_0 = 18$,*against* $H_A : \mu \neq 18$

The test statistic is $t = \dfrac{\bar{x} - \mu_0}{\sqrt{\dfrac{s^2}{n}}} = \dfrac{18.5 - 18}{\sqrt{\dfrac{95.9}{6}}} = 0.125$

$|t| < t_{.05,5} = 2.571$, H_0 is accepted. The average number of pharmacies per region is 18.

7. The sample size n is given by $n = \dfrac{z^2 \; v(x)}{d^2} = \dfrac{(1.96)^2 (6.35)^2}{(0.5)^2} = 97.57 \sim 98$

Here z is the tabulated value of normal distribution at 5% level = 1.96, d= margin of error.

8. The estimate of proportion of regions in which there are less than 18 pharmacies is given by P = a / n = 3/6 =0.5. Here a = number of regions in the sample in which there are less than 18 pharmacies = 3.

9. The estimated variance of p is given by

$v(p) = \dfrac{N - n}{N(n-1)} pq = \dfrac{28 - 6}{28(6-1)} 0.5x0.5 = 0.039$. Here q = 1 – p = 1 - 0.5 = 0.5

10. The sample size n is given by $n = \dfrac{z^2 \; pq}{d^2} = \dfrac{(1.96)^2 \, x \, 0.75 x 0.25}{(0.2)^2} = 18$

11. We need to test the null hypothesis $H_0 : P = P_0 = .6$ against $H_A : P \neq P_0$.

The test statistic is $z = \dfrac{p - P_0}{\sqrt{\dfrac{P_0 Q_0}{n}}} = \dfrac{0.5 - 0.6}{\sqrt{\dfrac{0.6 X 0.4}{6}}} = -0.5$ [$Q_0 = 1 - P_0 = 1 - 0.6 = 0.4$]

$|z| = 0.5 < 1.96$, H_0 is accepted. the proportion of regions in which there are less than 18 pharmacies is 0.6.

Sampling Distribution: The distribution of sample means or sample variances or any function of these is known as sampling distribution. Some distributions are as:

(i) Student's t – Distribution, **(ii)** Chi-square (χ^2) Distribution,

(iii) F – distribution [Distribution of Variance Ratio].

Student's t- Distribution and Chi-square (χ^2) Distribution:

Let $x_1, x_2, \ldots\ldots x_n$ *be n observations selected from* $N(\mu, \sigma^2)$. *Then, we have*

$\bar{x} = \frac{1}{n}\Sigma x$, *and* $s^2 = \frac{1}{n-1}[\Sigma x^2 - \frac{(\Sigma x)^2}{n}]$. *Here*

$t = \frac{\bar{x} - \mu}{\sqrt{\frac{s^2}{n}}}$ *is distributed as Student's t with* $n-1 d.f.$ *and* $\chi^2 = \frac{(n-1)\, s^2}{\sigma^2}$ *is distributed as chi – square with* $(n-1)\ d.f.$

$F = \frac{s_1^2}{s_2^2}$ *is distributed as* var*iance ratio with* $(n_1 - 1)$ *and* $(n_1 - 1)$ *and* $(n_2 - 1)\ d.f.$ *Here* s_1^2 *is the* var*iance of first sample and* s_2^2 *is the* var*iancer of* sec*ond sample.*

Let

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

This z is similar to t and it is distributed as Normal distribution with mean zero and variance 1. It is used if sample size n is big (≥ 30)) and / or σ is known. If σ is not known and sample is small (n< 30) , then t-statistic is used.

Sample Questions

1. Define sampling and simple random sampling with example.
2. Show, by an example, that mean of simple random sample is an unbiased estimate of population mean; sample variance is an unbiased estimate of population variance.
3. The data given below are the emails received by an organization in different days, x: 10,12,11,15, 3,10,13, 9, 6, 20, 6, 4, 11, 10, 12,15,25, 21, 9, 13,17, 8, 9, 15,11,13,10,7,11, 5,

i) Select 10 days using Simple Random Sampling (SRS) and estimate the average number of emails received per day . Also find 95% confidence interval for the average number of emails.

ii) Select 6 days by (a) systematic sampling and (b) circular systematic method

- Estimate the standard error of the estimated total number of emails using both sampling methods.
- Estimate the proportion of days which emails received is less than 9 along with its standard error from (a).

4. The number of miss calls received by a person in different days are randomly observed and are as, Number of miss calls: 5, 3, 6, 4, 2, 8, 2, 6, 5, 3. Where, N= 50

(i) Estimate average number of miss calls received per day along with the standard error of the estimated average.

(ii) Is the average 7 of number of miss calls received by a person in different days?

Table of Random Numbers

36518 36777 89116 05542 29705 83775 21564 81639 27973 62413 85652 62817 57881
46132 81380 75635 19428 88048 08747 20092 12615 35046 67753 69630 10883 13683
31841 77367 40791 97402 27569 90184 02338 39318 54936 34641 95525 86316 87384
84180 93793 64953 51472 65358 23701 75230 47200 78176 85248 90589 74567 22633
78435 37586 07015 98729 76703 16224 97661 79907 06611 26501 93389 92725 68158
41859 94198 37182 61345 88857 53204 86721 59613 67494 17292 94457 89520 77771
13019 07274 51068 93129 40386 51731 44254 66685 72835 01270 42523 45323 63481
82448 72430 29041 59208 95266 33978 70958 60017 39723 00606 17956 19024 15819
25432 96593 83112 96997 55340 80312 78839 09815 16887 22228 06206 54272 83516
69226 38655 03811 08342 47863 02743 11547 38250 58140 98470 24364 99797 73498
25837 68821 66426 20496 84843 18360 91252 99134 48931 99538 21160 09411 44659
38914 82707 24769 72026 56813 49336 71767 04474 32909 74162 50404 68562 14088
04070 60681 64290 26905 65617 76039 91657 71362 32246 49595 50663 47459 57072
01674 14751 28637 86980 11951 10479 41454 48527 53868 37846 85912 15156 00865
70294 35450 39982 79503 34382 43186 69890 63222 30110 56004 04879 05138 57476
73903 98066 52136 89925 50000 96334 30773 80571 31178 52799 41050 76298 43995
87789 56408 77107 88452 80975 03406 36114 64549 79244 82044 00202 45727 35709
92320 95929 58545 70699 07679 23296 03002 63885 54677 55745 52540 62154 33314
46391 60276 92061 43591 42118 73094 53608 58949 42927 90993 46795 05947 01934
67090 45063 84584 66022 48268 74971 94861 61749 61085 81758 89640 39437 90044
11666 99916 35165 29420 73213 15275 62532 47319 39842 62273 94980 23415 64668
40910 59068 04594 94576 51187 54796 17411 56123 66545 82163 61868 22752 40101
41169 37965 47578 92180 05257 19143 77486 02457 00985 31960 39033 44374 28352
76418